TEMPO: Physics-Informed Urban Flow Simulation via Velocity-Driven Latent Flow Matching

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Abstract

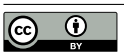
Probabilistic spatiotemporal generation of urban crowd flow is central to smart-city planning, yet existing generative approaches—most notably diffusion models—enforce physical regularity only through loss-based *soft penalties* and lack any architectural hard constraint. As a result, the generated density fields routinely create or destroy mass out of nowhere and exhibit physically implausible artifacts. To resolve this tension, we introduce **TEMPO**, the first framework to bring *Flow Matching* to urban crowd-flow generation. TEMPO embeds a *dual physical inductive bias* in the latent space: (1) a *structured velocity–density pipeline* that decodes the latent state into a continuous velocity field and advects density through a hard-coded discrete continuity equation, treating mass conservation as an architectural inductive bias rather than a loss penalty; and (2) a *State-Informed Prior* that anchors the generative starting point at the observed initial spatiotemporal state, substantially shortening the transport path and confining stochastic trajectories to the physically feasible region. On the large-scale real-world dataset **YJMob100K**, TEMPO matches state-of-the-art predictive accuracy while reducing the mass-conservation error by **55.0%**; ablations further show that the State-Informed Prior yields an additional **16.2%** reduction in mass error, confirming the effectiveness of coupling a spatial physical mechanism with the generative prior.

CCS Concepts

• **Computing methodologies** → **Machine learning**; *Continuous simulation*; • **Information systems** → **Spatial-temporal systems**.

Keywords

urban crowd flow simulation, human mobility, flow matching, physics-informed generative models, spatiotemporal generation, conditional generation



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1 Introduction

Spatiotemporal simulation and generation of urban crowd-flow density is a core capability for building smart-city infrastructure and for assessing the risk of extreme events. Because mobility systems are highly stochastic and high-dimensional, conventional deterministic regression models—e.g., convolutional [14] or graph-neural-network [12] approaches—produce only point estimates and cannot quantify the underlying spatiotemporal uncertainty. To address this, recent work has turned to generative frameworks (e.g., DiffSTG [11], KSTDiff [16]) that model probability distributions to capture the diversity of urban dynamics.

However, existing generative mobility models suffer from a systematic deficiency at the geographic and physical level. Real-world urban flow is strictly governed by *mass conservation*: any change in the density of a spatial cell must be driven by velocity-mediated transport between neighboring cells. Existing methods largely treat the underlying dynamics as a black box and impose physical regularity only as a *soft penalty* in the loss function [2]. During inference, stochastic perturbations easily breach such soft constraints, producing severe physical distortion and spurious creation or destruction of mass in the generated density field—undermining the reliability of the simulation for real-world geographic decision-making.

To resolve physical distortion *without sacrificing generative diversity*, we propose **TEMPO** (**T**emporal **E**volution via **M**ass-preserved **P**hysics **O**perator), the first framework to introduce a *Flow Matching* generative architecture to urban crowd-flow simulation, injecting spatial physical consistency into the forward generative process as a *structural inductive bias*. Our contributions are as follows:

(i) Structured velocity–density pipeline. Instead of generating density directly, the model captures the spatiotemporal trajectory in a latent space and decodes it into a dynamic velocity field; a hard-coded discrete continuity-equation operator then advects

density in the forward pass, embedding a conservation-aware physical inductive bias at the architectural level and reducing physically implausible density evolution.

(ii) State-Informed Prior. Since standard Flow Matching starts from uninformative Gaussian noise—yielding long transport paths that easily leave the physically feasible region—we anchor the generative start at the observed initial physical state to contract the trajectory, and adopt a *decoupled three-stage training strategy* that eliminates the train–inference distribution shift and maximizes the mass-conservation benefit.

2 Related Work

Generative mobility modeling. Traditional urban crowd-flow prediction relies on deterministic architectures [12, 14], while recent work adopts diffusion-based generative frameworks [11, 16] to capture uncertainty. Both, however, treat physical laws only as a soft-penalty loss [2], which stochastic perturbations at inference can readily violate. In contrast, TEMPO hard-codes the continuity equation into the architecture as a structural inductive bias, resolving the tension between physical fidelity and generative expressiveness.

Flow Matching and data-dependent priors. Flow Matching (FM) and rectified flow [6, 8] transport a source distribution to a target via a learned continuous-time velocity field, yielding nearly straight paths with lower training variance and faster sampling than diffusion—well suited to structured spatiotemporal generation. Standard FM, however, starts from an uninformative Gaussian prior, forcing a long path that readily leaves the physically feasible region; recent image-domain work (I^2SB [7], DDBM [15]) shows that data-dependent priors shorten this transport. TEMPO is the first to tailor this to the spatiotemporal setting, using the observed initial state as a *State-Informed Prior* that constrains mass conservation at the source without added inference bias.

3 Methodology

3.1 Problem Definition and Architecture Overview

Given an initial density frame $D_{t_0} \in \mathbb{R}^{H\times W}$, a POI map $P \in \mathbb{R}^{K\times H\times W}$, and a temporal index (hour, day-of-week), we learn the conditional distribution

$$p\big(D_{t_0+1:T} \mid D_{t_0},\, P,\, h,\, w\big), \tag{1}$$

to generate a crowd-flow density sequence over the future $T = 6$ time steps. As shown in Figure 1, the TEMPO framework comprises four core components: (1) a *spatial and contextual encoder* that compresses high-dimensional spatial and temporal features onto a continuous manifold; (2) *latent Flow Matching with the State-Informed Prior* (§3.3), which performs efficient conditional trajectory generation in the latent space; (3) a *causal Velocity Decoder* that maps the latent trajectory to a dynamic velocity field; and (4) *Physics-Guided Density Evolution* (§3.4).

3.2 Spatial Encoder

The vector-field sampling of Flow Matching relies critically on a continuous transport path between the source and target distributions; if the latent space contains discontinuous regions, the FM inference trajectory is prone to collapse. We therefore adopt a VAE [5] rather than a plain autoencoder, using the KL divergence to pull the latent features toward an isotropic Gaussian prior and thereby ensure global smoothness of the latent space. More importantly, urban crowd-flow simulation is highly sensitive to spatial hotspots and boundaries, where conventional spatial down-sampling erases fine-grained local peaks through a smoothing effect. Our Spatial Encoder therefore *preserves the original spatial resolution* during compression, so that the latent representation accurately perceives the location and spatial intensity of highly dynamic flow hotspots.

3.3 Latent Flow Matching and the State-Informed Prior

We adopt Optimal-Transport Conditional Flow Matching (OT-CFM) [10] to perform conditional generation in the latent space. Given a source sample x_0 and a target $x_1 = z$, the linear interpolation path is $x_t = (1-t)\,x_0 + t\,x_1$, and the training objective is

$$\mathcal{L}_{\mathrm{FM}} = \mathbb{E}_{t,x_0,x_1}\left\| v_\theta(x_t, t, c) - (x_1 - x_0) \right\|^2. \tag{2}$$

The vector field v_θ is parameterized by a 3D U-Net that conditions on both the spatial POI feature map (channel concatenation) and the global POI/temporal encoding (FiLM modulation), a dual-path design that preserves the positional and categorical information of POIs.

Compared with denoising diffusion probabilistic models (DDPM) [3], Flow Matching has straight transport paths and low training variance, making it better suited to high-fidelity spatiotemporal generation. Standard FM, however, samples from uninformative Gaussian noise, which in spatiotemporal flow simulation forces the model along a long transport trajectory that easily departs from the spatial physical constraints. To overcome this limitation, we propose the *State-Informed Prior*. Rather than pure Gaussian noise, we use the observed initial spatiotemporal state and, through a lightweight spatial encoder, sample a geographically informed conditional prior distribution as the transport origin. This design anchors FM's straight-line transport within the physically feasible region, markedly shortening the generative path and suppressing the mass distortion induced by stochastic noise.

3.4 Physics-Guided Density Evolution

Real-world urban flow is, at the macroscopic level, strictly governed by mass conservation. Unlike first-order models [4] that reduce travel speed to an algebraic function of local density, we adopt a *second-order dynamics* model [1] that better reflects the directional inertia of real pedestrians. The module comprises three core steps. **(1) Velocity-field generation:** the Velocity Decoder uses a causal 3D residual network to map the FM-generated latent trajectory and the initial density into an independently evolving dynamic velocity field V; its causal structure effectively prevents future-information leakage. **(2) Hard-coded physical constraint:** departing from conventional soft-penalty losses, we hard-code the geographic-space continuity-equation operator directly into the forward pass, driving density evolution through velocity-mediated transport between neighboring cells and promoting mass-consistent density evolution at the architectural level. **(3) Dynamic correction and**

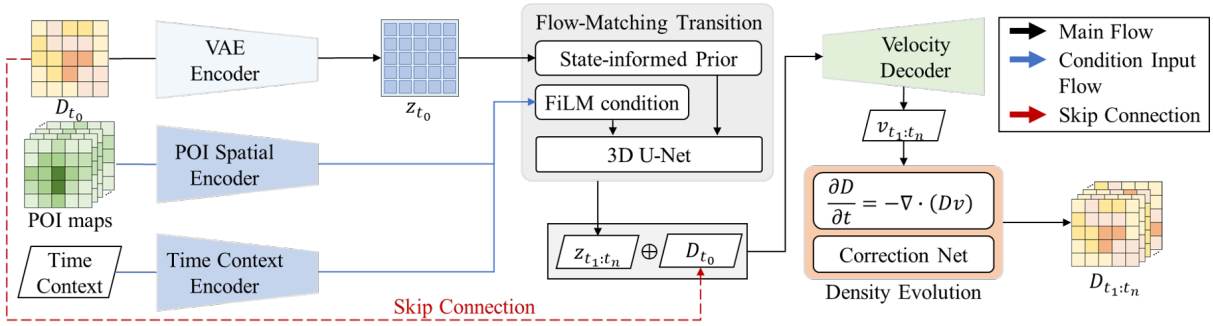


Figure 1: TEMPO architecture. A frozen VAE encoder maps D_{t_0} and the POI map to a structured latent; latent Flow Matching generates the trajectory from a State-Informed Prior anchored at D_{t_0}; a causal Velocity Decoder yields the velocity field V; and Physics-Guided Density Evolution advects density via a hard-coded continuity equation with a lightweight CorrectionNet. Shaded blocks mark our two contributions.

non-negativity: a small CNN (CorrectionNet) compensates for the gap between complex reality and the idealized physical equation. The final evolution is

$$D_{t+1} = \mathrm{ReLU}\Big(D_t - \Delta t\, \nabla\cdot\big(D_t V_t\big) + \mathrm{Correction}_\psi\big(D_t, V_t\big)\Big), \quad (3)$$

where the divergence term is computed with a fixed convolution kernel and ReLU strictly enforces the non-negativity bound on the predicted density. The velocity field $V_t = (V_t^x, V_t^y)$ carries the horizontal and vertical components on the spatial grid, and the divergence is obtained by convolving the two flux channels $D_t V_t^x$ and $D_t V_t^y$ with a pair of fixed 3×3 central-difference kernels ($[-\frac{1}{2}, 0, \frac{1}{2}]$ along the x- and y-axis, respectively) and summing the results, with zero padding at the spatial boundary. The kernels are registered as non-trainable buffers, so the operator introduces no parameters, and Eq. (3) is rolled out autoregressively over the T future steps with $\Delta t = 0.1$ in log-normalized density space.

3.5 Training Strategy

In sequence generation, an input-distribution shift between training and inference often induces severe *exposure bias* [9]. In our framework, the Velocity Decoder receives the VAE's sampled posterior during training but the ODE-integrated output of Flow Matching at inference. To eliminate this distribution shift and maximize the mass-conservation benefit, we propose a *decoupled three-stage training strategy*. **Stage 0 (spatial-manifold pretraining):** train the VAE with the ELBO loss to establish a smooth, global latent manifold, then freeze it so that subsequent gradients do not perturb the spatial features. **Stage 1 (Flow-Matching optimization):** freeze the encoder and train the 3D U-Net with the $\mathcal{L}_{\mathrm{FM}}$ objective, learning the vector field from the State-Informed Prior to the future latent trajectory. **Stage 2 (velocity and physics fine-tuning):** freeze the encoder and FM, and re-fine-tune the Velocity Decoder and CorrectionNet using the latent trajectories actually obtained by FM inference sampling; following the principle of *exactly matching the training distribution to the test distribution*, this stage maximally optimizes mass conservation.

4 Experiments

4.1 Experimental Setup

We validate our approach on the large-scale real-world trajectory dataset YJMob100K [13], which contains the mobility trajectories of 100,000 individuals together with point-of-interest (POI) data over 75 days in a metropolitan area of Japan. Missing values are imputed by linear interpolation, so that sampling gaps neither create nor destroy flow in the resulting density fields. The 75 POI categories are aggregated into 10 functional classes, and we model and evaluate in log-normalized space.

Evaluation spans three dimensions: (1) *point-prediction accuracy*, using MAE and RMSE; (2) *generative distributional quality*, using the Continuous Ranked Probability Score (CRPS), 50% calibration (Cal-50%), and Sample Div (across-sample standard deviation); and (3) *physical conservation*, via the *Mass (Conservation) Error*, which measures the mean relative error between the predicted and ground-truth total density over future time steps across the entire spatial map in log-normalized space:

$$\mathrm{Mass} = \frac{1}{T}\sum_{t=1}^{T}\left|\frac{\sum_{x,y} D_{t,x,y} - \sum_{x,y} D^{GT}_{t,x,y}}{\sum_{x,y} D^{GT}_{t,x,y}}\right|, \quad (4)$$

where $D_{t,x,y}$ is the model's predicted density.

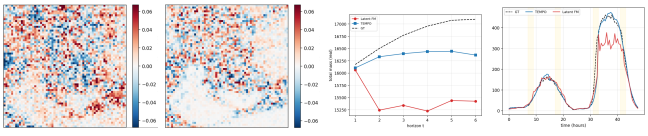
4.2 Comparison with Baselines

We compare against two deterministic predictors (ST-ResNet [14], Graph WaveNet [12]), a classical spatiotemporal diffusion model (DiffSTG [11]), and an unconstrained latent Flow-Matching baseline (Latent FM baseline [10]). Results are reported in Table 1.

TEMPO attains the best point accuracy (MAE, RMSE) and distributional fit (CRPS). Crucially, relative to the unconstrained Latent FM in the same latent space, it reduces the mass-conservation error by **55.0%**, demonstrating the value of hard-coding the continuity equation as a *structural inductive bias*. The diffusion baseline (DiffSTG) shows a much higher mass error (0.1560) and poorer point accuracy, reflecting how unconstrained black-box models decouple statistical correlation from physical mechanism. TEMPO is slightly conservative on calibration (Cal-50%) and sample diversity, revealing the trade-off between a hard physical constraint and generative spread.

Table 1: Main results on YJMob100K (test set; mean over 5 seeds). Lower is better for MAE, RMSE, Mass Error and CRPS; for Cal-50% values closer to 0.5 indicate better calibration; Sample Div reports generative spread and is not a better/worse axis; "–" marks metrics inapplicable to deterministic predictors; best per column in bold.

	Method	MAE	RMSE	Mass Error	CRPS	Sample Div	Cal-50%
Baselines	ST-ResNet	0.0200	0.0335	0.0293	–	–	–
	Graph WaveNet	0.0496	0.0648	0.1756	–	–	–
	DiffSTG	0.0225	0.0354	0.1560	0.0196	0.0212	0.2597
	Latent FM baseline	0.0112	0.0169	0.0253	0.0081	0.0159	0.5328
	TEMPO	**0.0093**	**0.0152**	**0.0114**	**0.0076**	0.0071	0.4230
(A) Physics ablation	– no CorrectionNet	0.0154	0.0274	0.1319	0.0137	0.0073	0.3010
	– no continuity physics	0.0164	0.0318	0.1427	0.0164	0.0000	0.4446
(B) Training-strategy ablation	– no Decoder fine-tune	0.0097	0.0161	0.0260	0.0080	0.0068	0.4153
(C) Prior-design ablation	Standard Gaussian prior	0.0093	0.0154	0.0136	0.0076	0.0072	0.3930



(a) (b) (c) (d)

Figure 2: Physical plausibility: TEMPO vs. unconstrained Latent FM. (a, b) Continuity residuals (same mass-conserving optical flow): Latent FM is mottled, TEMPO near zero. (c) Per-time-step total mass: TEMPO tracks ground truth smoothly, Latent FM fluctuates. (d) Density at one hotspot pixel over ∼2 days: TEMPO follows ground truth, Latent FM underestimates peaks (gold band: commuting peak).

To examine TEMPO's physical inductive bias, Figure 2 compares its spatial evolution with the unconstrained Latent FM. Spatially, panel (a) shows the Latent FM spuriously creating and destroying flow—leaving large unexplained residuals and global mass loss—whereas panel (b) shows TEMPO's residuals are markedly smaller, approaching zero in sparse regions. Temporally, panel (c) shows TEMPO's total mass stays stable and close to the ground truth, and panel (d) shows it tightly tracks multi-day evolution while the Latent FM underestimates peaks and fluctuates. Together, this conservation, local continuity, and temporal coherence confirm the effectiveness of the physical inductive bias.

4.3 Ablation Study

To isolate the contribution of each key design, we ablate the core physics module, the training strategy, and the State-Informed Prior; the lower half of Table 1 reports the results. **(1) Core physics module.** Removing CorrectionNet entirely raises MAE to 0.0154, confirming that the correction term is necessary; removing the continuity term drives the mass error sharply up to 0.1427, fully demonstrating the decisive contribution of the architectural physics constraint to conservation. **(2) Training strategy.** Omitting the Decoder fine-tuning inflates the mass error to 0.0260 (↑ 128.1%), confirming that eliminating the distribution shift between training (reparameterized noise) and inference (FM-sampled trajectories) substantially contributes to improved mass conservation. **(3) Prior design.** Reverting the State-Informed Prior to a standard Gaussian prior $\mathcal{N}(0, I)$ raises the mass error from 0.0114 to 0.0136 (↑ 19.3%), confirming that using the observed initial physical state as the prior markedly contracts the generative trajectory.

5 Conclusion

We present **TEMPO**, the first framework to bring Flow Matching to urban crowd-flow generation, incorporating a hard-coded continuity operator and a State-Informed Prior to substantially improve mass conservation. On YJMob100K, TEMPO matches state-of-the-art accuracy while substantially reducing mass error, resolving the tension between generative fidelity and diversity. Future work will balance the neural correction and hard-coded physics terms, extend the evaluation beyond YJMob100K to additional cities and to conservation-aware baselines, and extend the framework to non-routine scenarios such as large events or emergency evacuation to assess out-of-distribution generalization.

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